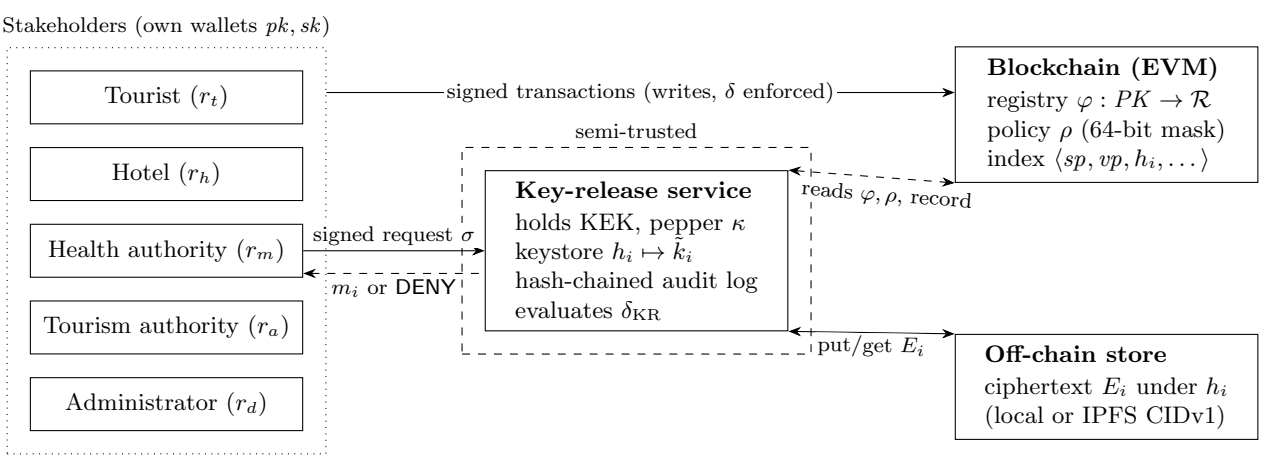

Stakeholders (own wallets pk, sk)
Tourist (r_t)
Hotel (r_h)
Health authority (r_m)
Tourism authority (r_a)
Administrator (r_d)
signed transactions (writes, δ enforced)
Blockchain (EVM)
registry $\varphi : PK \to \mathcal{R}$
policy ρ (64-bit mask)
index $\langle sp, vp, h_i, \dots \rangle$
semi-trusted
Key-release service
holds KEK, pepper κ
keystore $h_i \mapsto \tilde{k}_i$
hash-chained audit log
evaluates δ_{KR}
reads φ, ρ, record
signed request σ
m_i or DENY
put/get E_i
Off-chain store
ciphertext E_i under h_i
(local or IPFS CIDv1)